

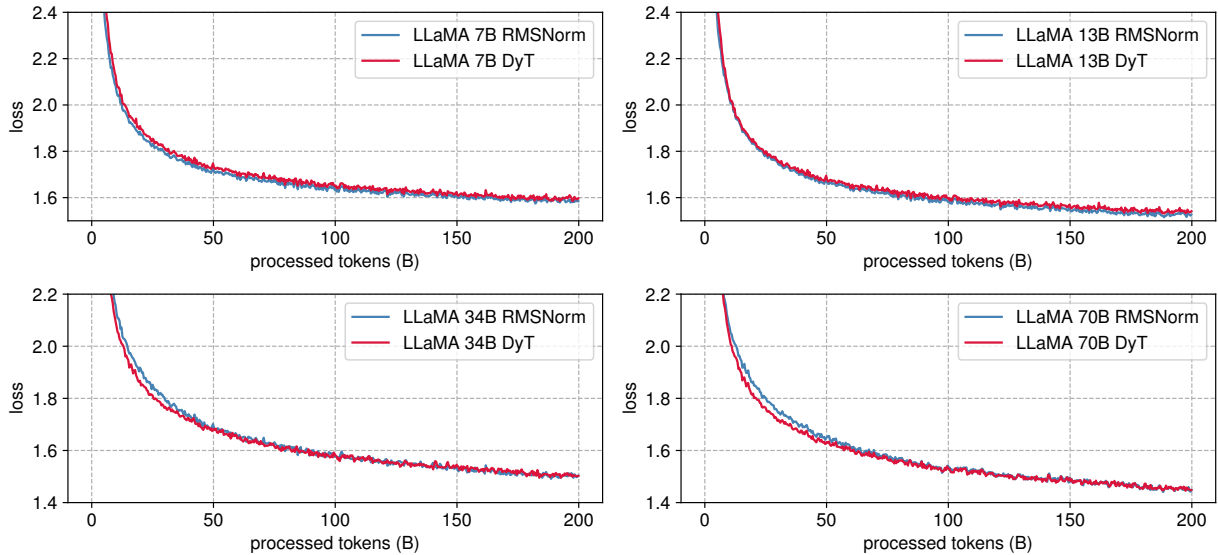


Figure 6 LLaMA pretraining loss. The loss curves of DyT and RMSNorm models are closely aligned across model sizes.

Large Language Models. We pretrain LLaMA 7B, 13B, 34B, and 70B models (Touvron et al., 2023a,b; Dubey et al., 2024) to assess DyT performance relative to RMSNorm (Zhang and Sennrich, 2019), the default normalization layer used in LLaMA. The models are trained on The Pile dataset (Gao et al., 2020) with 200B tokens, following the original recipe outlined in LLaMA (Touvron et al., 2023b). On LLaMA with DyT, we add a learnable scalar parameter after the initial embedding layer, and adjust the initial value of α , as detailed in Section 7. We report the loss value after training and also follow OpenLLaMA (Geng and Liu, 2023) to benchmark the models on 15 zero-shot tasks from lm-eval (Gao et al.). As shown in Table 4, DyT performs on par with RMSNorm across all four model sizes. Figure 6 illustrates the loss curves, demonstrating similar trends across all model sizes, with training losses closely aligned throughout training.

score / loss	RMSNorm	DyT	change
LLaMA 7B	0.513 / 1.59	0.513 / 1.60	- / $\uparrow 0.01$
LLaMA 13B	0.529 / 1.53	0.529 / 1.54	- / $\uparrow 0.01$
LLaMA 34B	0.536 / 1.50	0.536 / 1.50	- / -
LLaMA 70B	0.549 / 1.45	0.549 / 1.45	- / -

Table 4 Language models’ training loss and average performance with 15 zero-shot lm-eval tasks. DyT achieves a comparable zero-shot performance and training loss to RMSNorm.

Self-supervised learning in speech. We pretrain two wav2vec 2.0 Transformer models (Baevski et al., 2020) on the LibriSpeech dataset (Panayotov et al., 2015). We report the final validation loss in Table 5. We observe that DyT performs comparably to LN in both model sizes.

model	LN	DyT	change
wav2vec 2.0 Base	1.95	1.95	-
wav2vec 2.0 Large	1.92	1.91	$\downarrow 0.01$

Table 5 Speech pretraining validation loss on LibriSpeech. DyT performs comparably to LN for both wav2vec 2.0 models.

DNA sequence modeling. On the long-range DNA sequence modeling task, we pretrain the HyenaDNA model (Nguyen et al., 2024) and the Caduceus model (Schiff et al., 2024). The pretraining uses the human reference genome data from (GRCh38, 2013), and the evaluation is on GenomicBenchmarks (Grešová et al., 2023). The results are presented in Table 6. DyT maintains performance comparable to LN for this task.

model	LN	DyT	change
HyenaDNA (Nguyen et al., 2024)	85.2%	85.2%	-
Caduceus (Schiff et al., 2024)	86.9%	86.9%	-

Table 6 DNA classification accuracy on GenomicBenchmarks, averaged over each dataset in GenomicBenchmarks. DyT achieves comparable performance to LN.

6 Analysis

In this section, we begin with ablations on the effects of the tanh function and the learnable scalar α . We then analyze the values of α throughout and after training. Lastly, we present comparisons with previous methods that aim to remove normalization layers.

6.1 Ablations of tanh and α

To further investigate the role of tanh and α in DyT, we conduct experiments to evaluate the model’s performance when these components are altered or removed.

Replacing and removing tanh. We replace tanh in DyT layers with alternative squashing functions, specifically hardtanh and sigmoid (Figure 7), while keeping the learnable scalar α intact. Furthermore, we assess the impact of completely removing tanh by replacing it with the identity function while still retaining α . As shown in Table 7, the squashing function is essential for stable training. Using the identity function leads to unstable training and divergence, whereas squashing functions enable stable training. Among the squashing functions, tanh performs the best. This is possibly due to its smoothness and zero-centered properties.

model	identity	tanh	hardtanh	sigmoid
ViT-S	58.5% $\rightarrow$ failed	80.3%	79.9%	79.6%
ViT-B	61.0% $\rightarrow$ failed	82.5%	82.2%	81.6%

Table 7 ImageNet-1K classification accuracy with different squashing functions. All experiments follow the same training recipe as the original LN-based models. Squashing functions play a crucial role in preventing divergence, with tanh achieving the highest performance among the three functions. “ $\rightarrow$ failed” indicates that training diverged after some progress, with the preceding number representing the highest accuracy reached before divergence.

Removing α . Next, we evaluate the impact of removing the learnable α while retaining the squashing functions (tanh, hardtanh, and sigmoid). As shown in Table 8, removing α results in performance degradation across all squashing functions, highlighting the critical role of α in overall model performance.

model	tanh	hardtanh	sigmoid
without α	81.1%	80.7%	80.7%
with α	82.5%	82.2%	81.6%

Table 8 ImageNet-1K classification accuracy with ViT-B. All experiments follow the same training recipe as the original LN-based models. The learnable α is essential for enhancing model performance.

6.2 Values of α

During training. Our analysis reveals that the α closely tracks the $1/\text{std}$ of activations throughout training. As illustrated in the left panel of Figure 8, α first decrease and then increase during training, but always fluctuate consistently with the standard deviation of input activations. This supports the important role of α in maintaining activations within a suitable range, which leads to stable and effective training.

After training. Our further analysis of the final values of α in trained networks reveals a strong correlation with the $1/\text{std}$ of the input activations. As shown on the right panel of Figure 8, higher $1/\text{std}$ values generally correspond to larger α values, and vice versa. Additionally, we observe that deeper layers tend to have activations with larger standard deviations. This trend aligns with characteristics of deep residual networks, as shown in Brock et al. (2021a) for ConvNets, and Sun et al. (2025) for Transformers.

Both analyses suggest that α functions partially as a normalization mechanism by learning values approximating $1/\text{std}$ of the input activations. Unlike LN, which normalizes the activations per token, α normalizes the entire input activations collectively. Consequently, α alone cannot suppress extreme values in a non-linear fashion.

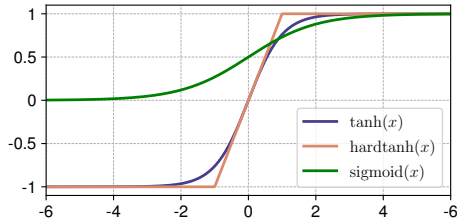


Figure 7 Curves of three squashing functions: $\tanh$, hardtanh , and sigmoid . All three functions squash inputs into a bounded range, but $\tanh(x)$ achieves the best performance when used in DyT layers. We suspect it is due to its smoothness and zero-centered properties.

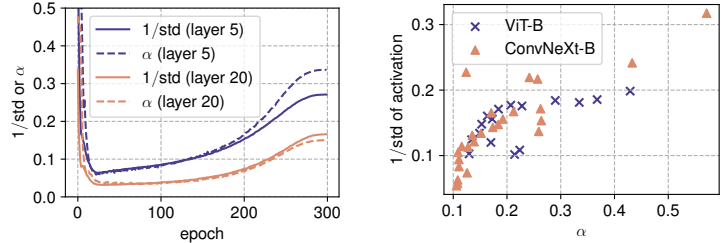


Figure 8 *Left:* For two selected DyT layers from the ViT-B model, we track α and the inverse of the standard deviation ($1/\text{std}$) of activations at the end of each epoch, observing that they evolve together during training. *Right:* We plot the final α values of two trained models, ViT-B and ConvNeXt-B, against the $1/\text{std}$ of the input activations, demonstrating a strong correlation between the two values.

6.3 Comparison with Other Methods

To further assess DyT’s effectiveness, we compare it with other methods that also enable training Transformers without normalization layers. These methods can be broadly categorized into initialization-based and weight-normalization-based methods. We consider two popular initialization-based methods, Fixup (Zhang et al., 2019; Huang et al., 2020) and SkipInit (De and Smith, 2020; Bachlechner et al., 2021). Both methods aim to mitigate training instabilities by adjusting the initial parameter values to prevent large gradients and activations at the start of training, thereby enabling stable learning without normalization layers. In contrast, weight-normalization-based methods impose constraints on network weights throughout training to maintain stable learning dynamics in the absence of normalization layers. We include one such method, σ Reparam (Zhai et al., 2023), which controls the spectral norm of the weights to promote stable learning.

model	LN	Fixup	SkipInit	σ Reparam	DyT
ViT-B	82.3%	77.2%	74.1%	82.5%	82.8%
ViT-L	83.1%	78.1%	75.6%	83.0%	83.6%
MAE ViT-B	83.2%	73.7%	73.1%	83.2%	83.7%
MAE ViT-L	85.5%	74.1%	74.0%	85.4%	85.8%

Table 9 Classification accuracy on ImageNet-1K. DyT consistently achieves superior performance over other methods.

Table 9 summarizes the results of two ViT-based tasks. We closely follow the original protocols outlined in their respective papers. However, we find that both initialization-based methods, Fixup and SkipInit, require significantly lower learning rates to prevent training divergence. To ensure a fair comparison, we conduct a simple learning rate search for all methods, including DyT. This produces results that differ from those reported in Section 5, where no hyperparameter is tuned. Overall, the results show that DyT consistently outperforms all other tested methods across different configurations.

7 Initialization of α

We find that tuning the initialization of α (denoted α_0) rarely leads to significant performance improvements. The only exception is LLM training, where careful tuning of α_0 yields noticeable performance gains. In this section, we detail our findings on the impact of α initialization.

7.1 Initialization of α for Non-LLM Models

Non-LLM models are relatively insensitive to α_0 . Figure 9 shows the effect of varying α_0 on validation performance across different tasks. All experiments follow the original setup and hyperparameters of their respective recipe. We observe that performance remains stable across a wide range of α_0 values, with values between 0.5 and 1.2 generally yielding good results. We observe that adjusting α_0 typically affects only the early stages of the training curves. The main exception is supervised ViT-L experiments, where training becomes unstable and diverges when α_0 exceeds 0.6. In such cases, reducing the learning rate restores stability, as detailed below.